

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Midterm
Duration: 1 hour 30 min

Semester: Summer 2025
Full Marks: 50

CSE 320: Data Communications

Answer the following questions [Total 3].
Figures in the right margin indicate marks.

SET A

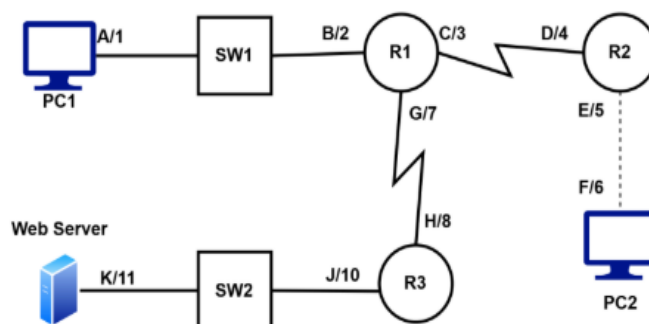
Name:	ID:	Section:
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1. [CO1] a) Suppose a dormitory has five buildings. Each of the buildings has three sections as follows: library, work station and break room. Each section has nodes as follows: four nodes for the library, 6 nodes for the work station and 2 nodes for the break room. Now, **draw** the hybrid topologies according to the following configuration, **labelling** the sections: [3+2]

- The nodes in each section are connected to the neighboring nodes only.
- The buildings have a dedicated connection to all other buildings.
- The sections are connected directly to a device which is a central point of failure. Meaning if it goes offline, then all the connected devices go offline.

Based on the above topology, calculate the total number of connection links.

- b) A web server using HTTPS (port number 443) has sent a response packet to a client PC1(port number 50000). The packet is currently at router R3, moving towards PC1. **Identify** the source and destination Port, IP, and MAC Addresses of this response packet. Consider the alphabets as physical addresses and the numbers as logical addresses (given beside the devices in the figure). [3]



- c) Imagine you are a network engineer working for a company that needs to send a message from Computer A to Computer B over a network. The message is: **“Hello, OSI!”** [4+2]

Your task is to break down how this message is processed at each layer of the OSI model.

You need to:

- For each layer, **identify** the specific name of the data unit and the type of addressing used at that layer.
- Illustrate** how headers and trailers are added as the data moves through the layers.

2. [CO2] a) A video streaming company is designing a network to stream 1080p HD video (1920x1080 pixels, 60 fps, 24-bit color) to users in a rural area. The network uses a wireless communication channel, which introduces some noise due to interference and long-range transmission. To ensure faster transmission, the video's raw bitrate will be reduced by a factor of 120 before final delivery. [4+3+3]

Wireless Channel Specs:

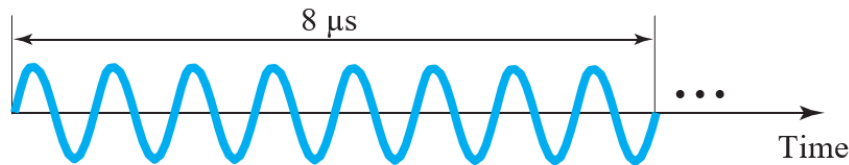
- Channel Bandwidth = 5 MHz
- Signal-to-Noise ratio = 20 dB

Hint: FPS means frames per second; each frame has the given resolution, and each pixel in that resolution uses the given color depth in bits.

You as an engineer need to determine:

- What is the **bitrate** required to stream the video considering the compressing?
 - What is the number of **signal levels** required for the stream assuming a noiseless channel. **How** many bits do you need to send per signal.
 - What is the maximum **achievable bitrate** over the actual noisy channel? Can the video stream be transmitted reliably over the current channel?
- b) If the signal-to-noise ratio (SNR) is close to zero for a communication channel, what will be the achievable capacity of that channel? For efficient communication, which is desirable: high SNR or low SNR? **Explain** your reasoning. [1+2]

- c) [3]



In the above Figure, **determine** the period and frequency of the analog signal. Also **show** the frequency domain representation of the signal if the maximum amplitude is 20V.

- d) **Distinguish** between lowpass channel and bandpass channel. [1+3]

If a signal at the beginning of a cable with **-0.45 dB/km** (the loss in a cable is usually defined in decibels per kilometer) has a power of 3.6 mW, **calculate** the power of the signal at 8.2 km.

3. [CO2] a) Suppose you are applying 3B/5B block coding in your network system. To get certain advantages, you are following some rules to map the 3-bit combinations with the 5-bit combinations. Observe the below mapping table - [3+1]

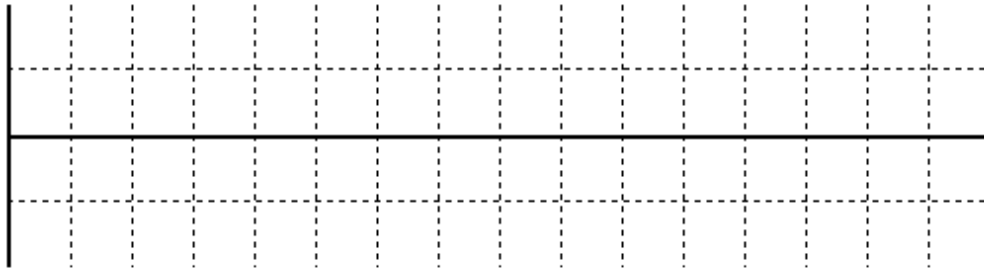
- **Write** down the rule/s that are applied to make a 5-bit codeword valid in the given table.
- If you are transferring the data at a speed of 100 kbps as a sender, then **what** will be the **transmission speed** after applying the 3B/5B substitution?

3-bit input	5-bit codeword
000	10110
001	11010
010	01011

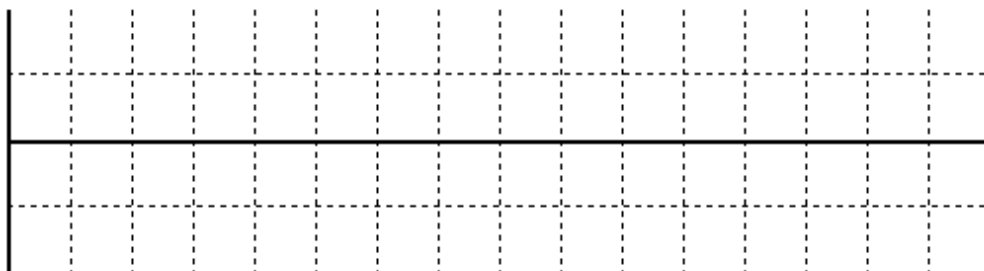
011	01110
100	10011
101	10101
110	11001
111	11110

- b) The NRZ-I scheme does not solve the problem of DC components. But, if the DC component problem is unacceptable, then **what scheme** can be used alternatively, which holds $r = 1$ (ratio of data element and signal element)? [1+2]

For the data stream 011000001000000, **apply** the line coding scheme that you are choosing.



- c) Now solve the long sequence of 0's problem in the above data stream given in 3(b) using a suitable method, keeping $r = 1$ (ratio of data element and signal element). [3]



- d) The following table depicts a sampled low pass analog signal with a bandwidth of 200 KHz. [4+2]

Time	Amplitude (v)
0	10.5
1	-18.3
2	60.5
3	-39.8

By applying the concept of **Pulse Code Modulation** for analog to digital conversion, **show** the normalized quantized values and encoded words for the given amplitudes at different time stamps. The sampling amplitudes are between -33V to +63V and there will be 8 levels in total.

